Question

Now there is a cubic crystal $\mathbf{A}$, which contains a cuboctahedral ($4^6 3^8$) polyhedron atomic cluster $\mathbf{Z}$ formed by boron atoms. The structure of $\mathbf{A}$ can be regarded as $\mathbf{Z}$ forming a NaCl-type structure with metal atom $\mathbf{X}$. It is known that all adjacent $\mathbf{Z}$ are connected by B-B bonds, and their bond lengths are all 169.2pm; all B-B bonds inside $\mathbf{Z}$ have bond lengths of 178.8pm. The theoretical density of the crystal is $5.908 \mathrm{g/cm^3}$. The crystal can also be regarded as the result of $\mathbf{X}$ atoms being encapsulated by another atomic cluster $\mathbf{Y}$.

Then the following statements are correct:

A. All other options are incorrect

B. The ratio of two elements in the chemical formula of $\mathbf{A}$ is 1:8.

C. The angle formed by the triangular face in $\mathbf{Z}$ and the (110) crystal plane is unique.

D. The coordination number of metal $\mathbf{X}$ in $\mathbf{Z}$ is 16.

E. $\mathbf{A}$ unit cell body diagonal length is greater than 1300 pm.

F. $\mathbf{X}$ The mass number of the stable nuclide produced finally by the natural radioactive series to which the most abundant natural isotope belongs is 208.

G. The structure of $\mathbf{Y}$ can be represented by the symbol $4^6 6^8$.

Answer

Correct Answer: G

Detailed Explanation

Basic Unit $\mathbf{Z}$:

A cuboctahedral ($4^6 3^8$) atomic cluster composed of boron atoms (B). The cuboctahedron has 12 vertices, 8 equilateral triangular faces, and 6 square faces. Therefore, $\mathbf{Z}$ is a B_{12} atomic cluster.

CHECKPOINT 1 PTS

The chemical formula of $\mathbf{Z}$ is B_{12}

Crystal Structure:

The structure of $\mathbf{A}$ can be regarded as $\mathbf{Z}$ and the metal atom $\mathbf{X}$ forming a NaCl-type structure. In the NaCl structure, the stoichiometric ratio of cations and anions (or units) is 1:1. This means that the number ratio of $\mathbf{Z}$ to $\mathbf{X}$ is 1:1, so the chemical formula of $\mathbf{A}$ should be $\mathbf{X}B_{12}$, with a ratio of the two elements being 1:12. Option B is incorrect.

CHECKPOINT 1 PTS

The ratio of the two elements in $\mathbf{A}$ is 1:12

According to the cubic crystal symmetry, the six quadrilateral faces in the atomic cluster $\mathbf{Z}$ are perpendicular to the a, b, and c axes, respectively. Therefore, it is easy to have two angles for the quadrilateral faces, one perpendicular to the (110) crystal plane and the other non-perpendicular. Option C is incorrect.

CHECKPOINT 1 PTS

The triangular faces in $\mathbf{Z}$ form 2 types of angles with the [110] crystal plane

According to the description, there are 6 $\mathbf{Z}$ clusters coordinating around $\mathbf{X}$. Each cluster uses a quadrilateral face for coordination, so the coordination number of $\mathbf{X}$ is $4*6=24$. Option D is incorrect.

CHECKPOINT 1 PTS

The coordination number of the metal $\mathbf{X}$ in $\mathbf{Z}$ is 24

The lattice parameter a is determined by $\sqrt{2}$ times the distance between the centers of two adjacent B_{12} clusters, and the distance between the centers of two adjacent B_{12} clusters is equal to the sum of the B-B bond length within a cluster and the B-B bond length between two clusters. Therefore, the lattice parameter of $\mathbf{Z}$ is $a=169.2*\sqrt{2}+178.8*\sqrt{2}*2=745.0pm$, so the diagonal length is $745.0*\sqrt{3}=1290.3pm$. Option E is incorrect.

CHECKPOINT 1 PTS

The body diagonal length of the $\mathbf{A}$ unit cell is 1290 pm.

The body diagonal length of the unit cell is less than 1300 pm.

$\mathbf{X}$ Element Deduction:

Use the density formula to calculate the molar mass of $\mathbf{A}$. From $\rho = \frac{Z \cdot M}{N_A \cdot V}$, we get $M_{\mathbf{X}B_{12}} = \frac{\rho \cdot N_A \cdot V}{Z} = \frac{5.908 * N_A * 7.450^3 * 10^{-24}}{4} = 367.8 g/\mathrm{mol}$, thus $M_{\mathbf{X}} = 367.8 - 12 \cdot M_{\mathbf{B}} = 238.1$, $\mathbf{X}$ is the element U.

CHECKPOINT 1 PTS

$\mathbf{X}$ is the element uranium (U)

The mass number of the stable nuclide finally produced by the natural radioactive series of its most abundant natural isotope, $^{238}\mathrm{U}$, is 206 (i.e., ^{206}Pb). Option F is incorrect.

CHECKPOINT 1 PTS

The mass number of the stable nuclide finally produced by $^{238}\mathrm{U}$ is 206

$\mathbf{Y}$ Structure Deduction

According to the coordination mode, there are 6 quadrilateral faces in $\mathbf{Y}$. These six faces can be regarded as being on the six vertices of an octahedron. From this, it can be seen that the three-dimensional structure is a truncated octahedron. Therefore, $\mathbf{Y}$ has 6 quadrilateral faces and 8 hexagonal faces, which can be represented by the symbol $4^6 6^8$. Option G is correct.

CHECKPOINT 1 PTS

The structure of $\mathbf{Y}$ can be represented by the symbol $4^6 6^8$.

Therefore, the correct option is: G